

Cataract 13 With Adult i Phenotype (CTRCT13): A Comprehensive Disease Report

Disease: Cataract 13 With Adult I Phenotype **MONDO ID:** MONDO:0007289 · **OMIM:** 116700 · **Gene:** *GCNT2* (6p24.2, OMIM 600429) **Category:** Mendelian (autosomal recessive) **Report date:** 2026-09-09 · **Evidence base:*** human clinical/genetic studies, biochemical/enzymatic assays, and comparative glycobiology. No patient-level dataset was supplied; this is a literature-derived synthesis.

Summary

Cataract 13 with adult i phenotype (CTRCT13) is a rare Mendelian disorder in which bilateral congenital cataract co-occurs with the benign **adult "i" blood group phenotype**, both arising from biallelic loss-of-function of a single gene, *GCNT2* (glucosaminyl (N-acetyl) transferase 2), on chromosome 6p24. *GCNT2* is the **I-branching β -1,6-N-acetylglucosaminyltransferase** (EC 2.4.1.150), the enzyme that converts the linear fetal poly-N-acetyllactosamine "i" antigen into the branched adult "I" antigen. When enzyme activity is lost, red-cell surface glycans remain in the linear "i" form (producing the serologically detectable adult i phenotype), and lens transparency is disrupted (producing congenital cataract).

The central mechanistic insight of this investigation is that the coupling between the blood-group trait and the cataract is only **partial**, and this is explained by the gene's **three-isoform architecture**. The human I locus expresses three transcripts — *GCNT2A*, *GCNT2B*, and *GCNT2C* — that share exons 2 and 3 but differ in exon 1. The **erythroid isoform (GCNT2C)** governs the blood-group phenotype, whereas a **lens-relevant isoform (GCNT2B, and the shared exons)** governs cataract. Mutations that hit the shared exons or delete all isoforms cause **both** phenotypes; mutations restricted to exon 1C cause the **adult i blood group without cataract**. This isoform logic resolves decades of clinical observation that the i–cataract association is strong in East Asian (Japanese) families but weak in White populations.

The allelic spectrum comprises nonsense and frameshift point mutations, activity-abolishing missense variants, and — a recurrent theme — **large Alu/SINE-mediated structural deletions** spanning one or more exons, reflecting the repeat-rich, rearrangement-prone nature of the 6p24 locus. Management is entirely **surgical and supportive** (early cataract extraction with amblyopia and refractive rehabilitation); life expectancy is normal, and no disease-modifying therapy targeting the underlying glycosylation defect exists. This report compiles the disease overview, genetics, mechanism, phenotypes, diagnostics, prognosis, treatment, and comparative/model-organism context, with primary-literature citations throughout.

Key Findings

Finding 1 — CTRCT13 is caused by biallelic *GCNT2* mutations

Autosomal-recessive congenital cataract with the adult i phenotype maps to *GCNT2* on chromosome 6p24 and is caused by **biallelic loss-of-function** of the gene. Homozygosity mapping in a consanguineous Pakistani family linked the recessive cataract to a 3.03 Mb locus on 6p24 that contains *GCNT2*; within it, a **homozygous 93 kb Alu-mediated deletion** removing exons 1B, 1C, 2, and 3 segregated with both cataract and the adult i blood group ([PMID: 21761136](#)). This paper states plainly that "*GCNT2* encodes glucosaminyl (N-acetyl) transferase 2, an enzyme responsible for the formation of the blood group I antigen," establishing both the gene and its enzymatic role in the disease.

Independent confirmation came from linkage and mutation analysis in four Arab families, where the locus was mapped with a maximum two-point **LOD score of 8.75** ($\theta = 0.019$ at marker D6S470) and a homozygous G→A substitution in exon 2 was identified ([PMID: 15161861](#)). That change "*revealed in these families a homozygous G→A substitution in base 58 of exon-2, resulting in the formation of premature stop codons W328X, W326X, and W328X, of the GCNT2A, -B, and -C isoforms.*" Because the premature stop lands in a shared exon, it truncates **all three isoforms simultaneously**, producing both the cataract and the adult i phenotype. Together these two studies — spanning distinct populations and distinct mutational classes (structural deletion vs. nonsense point mutation) — firmly establish *GCNT2* as the causal gene.

Ontology anchors: Gene *GCNT2* (HGNC:4204); MONDO:0007289; OMIM 116700; disease process GO:0006486 (protein glycosylation).

Finding 2 — Isoform-specific architecture explains the partial coupling of cataract and adult i

The human I locus expresses **three transcripts**, GCNT2A/B/C, that *"have different exon 1, but identical exons 2 and 3"* (PMID: 12424189). This modular design is the key to genotype–phenotype correlation:

- **GCNT2C** drives erythrocyte I-antigen expression, i.e., the **adult i blood group**.
- **GCNT2B** is the isoform associated with **lens function** (cataract) (PMID: 27609212).

Mutations affecting the **shared exons 2/3**, or deletions removing all isoforms, therefore cause **both** cataract and adult i. Conversely, a mutation restricted to **exon 1C** — such as the IGnTC 243T>A (Asn81Lys) variant — causes the **adult i blood group without cataract** (PMID: 17076854), a report that describes *"molecular genetic analysis of a Taiwanese person with the adult i phenotype but without congenital cataracts."* This is the clean genetic dissociation that proves the two traits are governed by different transcripts of the same gene.

A complementary case supports the mirror-image inference for the cataract-relevant transcript: a proband with congenital cataract carried a truncating variant in **exon 1B** (NM_001491.3: c.760dup p.H254Pfs*2) in trans with a 75 kb deletion of exons 1B/1C, implicating exon 1B-containing transcripts as clinically relevant for the cataract phenotype (PMID: 41813616).

Isoform → phenotype map:

GCNT2 isoform	Distinguishing exon	Tissue role	Phenotype when lost
GCNT2A	exon 1A	ubiquitous	contributes to systemic i
GCNT2B	exon 1B	lens	congenital cataract
GCNT2C	exon 1C	erythrocyte	adult i blood group
shared exons 2/3	—	all tissues	both cataract + adult i

Finding 3 — The adult i-cataract association is strongest in East Asians and weaker in Whites

The strength of the clinical coupling is **population-dependent**. In Japanese families, Ogata (1979) found that among 18 phenotype-i individuals across 10 families, **17 had congenital cataract**, whereas *"cataract was not found in any of the 45 phenotype I members in these families"* (PMID: 432928) — a near-complete co-segregation.

By contrast, in White populations the association is much weaker: only **3 of 10** White i-phenotype individuals had congenital cataract, and **0 of 31** White congenital-cataract patients had the i phenotype (PMID: 3799539), which reports *"three of ten people have congenital cataracts... the authors studied the blood of 31 white patients with congenital cataracts and found no patients with the i phenotype."* Combined with the isoform architecture of Finding 2, this pattern is best explained by **population-specific allelic effects**: East Asian pedigrees more often carry variants affecting shared exons (coupling the two traits), whereas White pedigrees more often carry isoform-restricted alleles that decouple them.

Population	i-individuals with cataract	Cataract patients with i phenotype	Association strength	Reference
Japanese	17/18	— (0/45 phenotype-I relatives had cataract)	Strong	PMID: 432928
White	3/10	0/31	Weak	PMID: 3799539

Finding 4 — Large Alu/SINE-mediated deletions are a recurrent mutational mechanism

Structural deletions are not incidental but a **major, recurrent class** of pathogenic *GCNT2* alleles, driven by the repeat-rich local genome. The locus *"is rich in Short INterspersed Elements (SINE repeats) and thus likely prone to genomic rearrangements"* (PMID: 21761136). Documented structural alleles include:

Deletion	Extent / breakpoint feature	Reference
93 kb (exons 1B, 1C, 2, 3)	Alu-mediated	PMID: 21761136
~98 kb at 6p24.3	Alu-mediated NHEJ; removes GCNT2 A/B first exons + intergenic region toward <i>TFAP2A</i>	PMID: 27609212
~190 kb insertion/deletion	large indel; PCR of coding exons failed in affected	PMID: 27936067
~75 kb (exons 1B/1C)	copy-number deletion in trans with exon-1B frameshift	PMID: 41813616

The ~190 kb allele was found in a large Pakistani pedigree with **LOD 5.78** at 6p24, where *"PCR amplifications of the coding exons of GCNT2 failed in individuals with arCC, and whole-exome data analysis revealed a large deletion on chromosome 6p in the region harboring GCNT2"* (PMID: 27936067). The practical diagnostic implication is important: **standard exon-based sequencing can miss the causal allele**, and failed PCR amplification of *GCNT2* exons should itself raise suspicion of a structural deletion requiring CNV/deletion-aware analysis (WES CNV calling, MLPA, or chromosomal microarray).

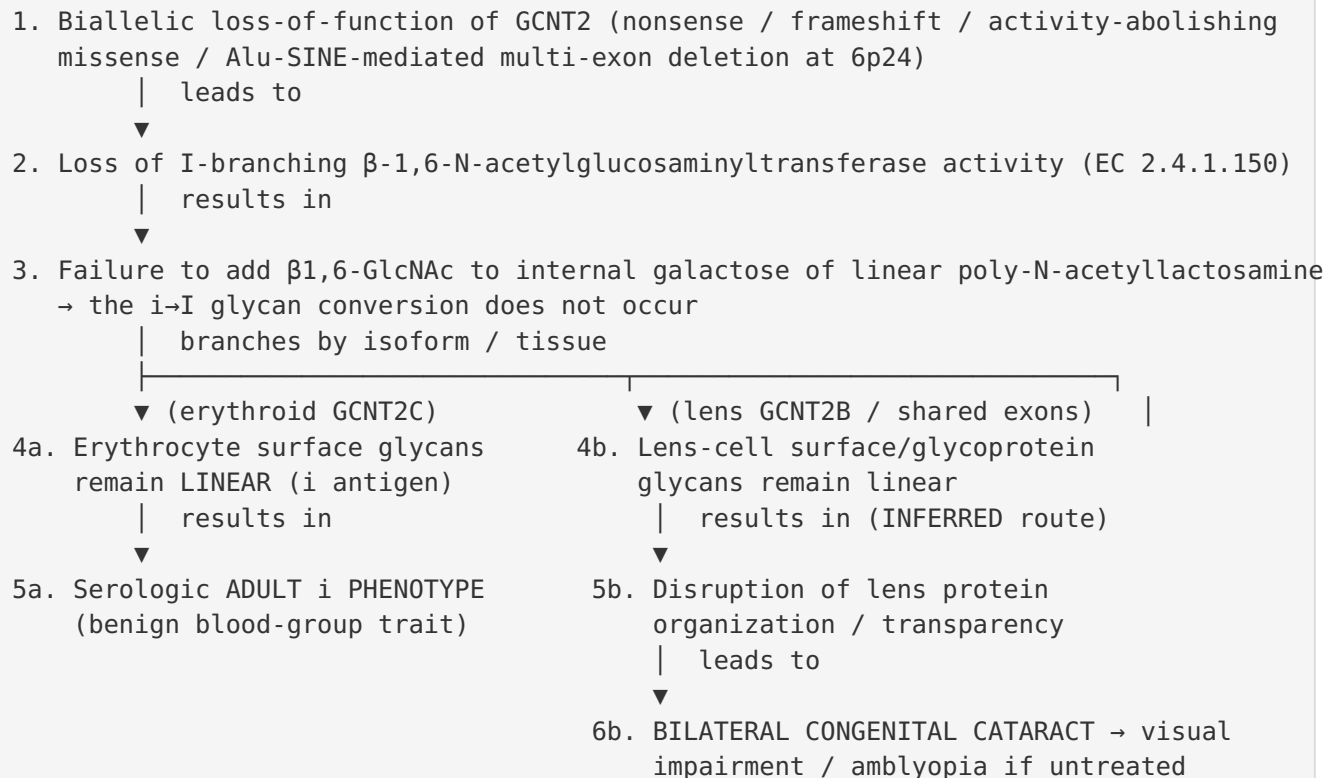
Finding 5 — Biochemical mechanism: GCNT2 catalyzes β 1,6-GlcNAc branching (i→I conversion)

GCNT2 is the **I-branching β -1,6-N-acetylglucosaminyltransferase (EC 2.4.1.150)**. It adds a β 1,6-linked GlcNAc to the internal galactose of linear poly-N-acetyllactosamine (polyLacNAc), converting the linear "i" antigen into the branched "I" antigen: *"Conversion of the i to the I structure requires I-branching beta-1,6-N-acetylglucosaminyltransferase activity"* (PMID: 12424189; see also PMID: 11739194). The enzyme cooperates with **β 3GnT2** to build and extend polyLacNAc chains — *"the β 1,6-branching glycosyltransferase GCNT2"* (PMID: 24105809).

When GCNT2 activity is lost, polyLacNAc remains **linear** on erythrocytes (the adult i phenotype) and, via the lens-relevant isoform, lens transparency is disrupted, yielding cataract. The **initiating lesion is biallelic loss of function**; however, the precise lens glycoprotein target and the biophysical route from an unbranched glycan to lens opacity **remain inferred rather than experimentally demonstrated**.

Mechanistic Model / Interpretation

Ordered causal chain



Upstream vs downstream. The upstream, proven event is loss of enzymatic branching activity. The **branch point** is tissue/isoform-specific expression: GCNT2C loss produces the blood-group readout; GCNT2B/shared-exon loss produces the cataract. Step 5b (how unbranched glycans translate biophysically into lens opacity) is the least characterized link — it is a reasonable inference from the enzyme's substrate chemistry but has not been mechanistically demonstrated in lens tissue.

Why the two traits are only partially coupled. Because the erythroid and lens phenotypes are driven by **different first exons of the same gene**, an allele's phenotypic footprint depends on which transcripts it disables. Shared-exon lesions (nonsense in exon 2/3; whole-gene deletions) hit everything → both traits. Isoform-restricted lesions (exon 1C only) → blood group only; (exon 1B only) → cataract-relevant transcript implicated. This single principle reconciles the East Asian vs. White epidemiological discrepancy (Finding 3): different populations carry different mixes of shared-exon vs. isoform-restricted alleles.

Cell types and processes. Primary cell types: **lens fiber cells** and **lens epithelial cells** (cataract) and **erythrocytes** (blood group). Core biological process: **protein glycosylation / polyLacNAc biosynthesis** in the Golgi apparatus.

Suggested ontology terms: - **GO (process):** GO:0006486 protein glycosylation; GO:0008375 acetylglucosaminyltransferase activity; GO:0047253 N-acetyllactosaminide β -1,6-N-acetylglucosaminyltransferase activity. - **GO (component):** GO:0000139 Golgi membrane. - **CL (cell types):** CL:0000362 epithelial cell (lens epithelium); CL:0011004 lens fiber cell; CL:0000232 erythrocyte. - **UBERON:** UBERON:0000965 lens of camera-type eye; UBERON:0000970 eye. - **CHEBI:** CHEBI:506227 N-acetyllactosamine; CHEBI:60152 N-acetyl-D-glucosamine (context).

Section-by-Section Report

1. Disease Information

- **Overview.** CTRCT13 is a rare autosomal-recessive Mendelian disorder combining **bilateral congenital cataract** with the benign **adult i blood group**. Both features stem from biallelic loss-of-function of *GCNT2*. Some individuals present cataract without the detectable i phenotype and vice versa, depending on the affected isoform.
- **Key identifiers.** OMIM **116700**; MONDO **0007289**; Gene *GCNT2* (HGNC:4204; NCBI Gene 2651; OMIM 600429). *ICD-10 maps broadly to congenital lens malformations (Q12.0** congenital cataract). MeSH concepts: "Cataract," "Blood Group Antigens." Orphanet groups this under early-onset/congenital genetic cataract entities.
- **Synonyms / alternative names.** "Cataract, congenital, with adult i phenotype"; "Adult i phenotype with congenital cataract"; "GCNT2-related congenital cataract"; historically linked to the "I/i blood group" and the "IGnT" gene nomenclature.
- **Information source.** Disease-level and family-based (aggregated pedigree/case reports and OMIM/ClinVar curation), **not** EHR-derived population cohorts.

2. Etiology

- **Causal factor.** Purely **genetic** — biallelic loss-of-function of *GCNT2*. No infectious or environmental cause. (Note: age-related cataract in the general population is multifactorial, but **CTRCT13 specifically is monogenic.**)

- **Genetic risk factors.** The disease requires two pathogenic *GCNT2* alleles. Reported classes: nonsense (e.g., W328X/W326X/W328X across isoforms, [PMID: 15161861](#)); frameshift (e.g., c.760dup p.H254Pfs2, [PMID: 41813616](#)); activity-abolishing missense (e.g., Gly348Glu, Arg383His in IGnT, [PMID: 11739194](#)); and large *Alu*/*SINE*-mediated deletions (Finding 4). *Consanguinity** is a major contributing factor — most pedigrees are consanguineous (Pakistani, Arab, Japanese).
- **Environmental risk / protective factors.** None established for the monogenic entity. No protective alleles identified.
- **Gene–environment interactions.** Not applicable / none demonstrated.

3. Phenotypes

Phenotype	Type	Onset	Severity / course	Frequency	Suggested HPO
Bilateral congenital cataract	clinical sign / physical	Congenital	Variable morphology; static lens opacity but visually progressive if untreated	Core feature (near-universal in shared-exon genotypes)	HP:0000519 (congenital cataract); HP:0000518 (cataract)
Adult i blood group	laboratory abnormality	Detectable lifelong (fails to convert i→I after infancy)	Benign, non-progressive	Present in blood-group-affecting genotypes	HP:0032324 (abnormal erythrocyte morphology, related); blood-group trait
Visual impairment / amblyopia (if untreated)	symptom	Infancy	Preventable with early surgery	Secondary	HP:0000505 (visual impairment); HP:0000646 (amblyopia)
Nystagmus (with dense bilateral congenital cataract)	clinical sign	Infancy	Secondary to deprivation	Variable	HP:0000639 (nystagmus)

- **Quality-of-life impact.** The **adult i phenotype is clinically benign** (relevant mainly for transfusion/serology). The cataract, if untreated, causes **deprivation amblyopia** and childhood blindness, with major QoL consequences; timely surgery largely restores function.

4. Genetic / Molecular Information

- **Causal gene.** *GCNT2* (glucosaminyl (N-acetyl) transferase 2 / I-branching enzyme; alias IGnT), 6p24.2, HGNC:4204, NCBI Gene 2651, OMIM *600429.

- **Variant classification (ACMG/AMP).** Nonsense, frameshift, and multi-exon deletions are typically **pathogenic**; activity-abolishing missense variants are pathogenic when supported by functional assays. Truncations in the clinically relevant exon 1B *"should likely receive strong pro-pathogenic weight"* (PMID: 41813616).
- **Variant types.** Missense (Gly348Glu, Arg383His, Asn81Lys), nonsense (W326X/W328X), frameshift (H254Pfs2), *and structural deletions** (75–190 kb).
- **Allele frequency.** Pathogenic alleles are **rare**; the disease is seen mainly in consanguineous pedigrees. Precise gnomAD frequencies were not compiled.
- **Origin. Germline**, biallelic (autosomal recessive). No somatic disease role.
- **Functional consequence. Loss of function** (loss of β 1,6-branching activity); no gain-of-function or dominant-negative mechanism.
- **Modifier genes / epigenetics.** None established. Note the *GCNT2–TFAP2A* intergenic region is deletion-prone; large deletions extending toward *TFAP2A* raise a theoretical concern for overlap with Branchio-Oculo-Facial Syndrome features, though patients generally do not meet BOFS criteria (PMID: 27609212).
- **Chromosomal abnormalities.** Recurrent **Alu/SINE-mediated microdeletions** at 6p24 (Finding 4).

5. Environmental Information

Not applicable. CTRCT13 is monogenic; **no** environmental toxins, lifestyle factors, or infectious agents are implicated in causing the disease.

6. Mechanism / Pathophysiology

See the **Mechanistic Model** section above for the ordered causal chain, branch logic, cell types, and ontology terms. In brief: *GCNT2* loss → no i→I branching of polyLacNAc → linear glycans on erythrocytes (adult i) and lens cells (cataract, via the biophysically inferred route). *GCNT2* cooperates with β 3GnT2 in polyLacNAc synthesis (PMID: 24105809). The reaction is EC 2.4.1.150 (PMID: 12424189).

7. Anatomical Structures Affected

- **Organ level.** Primary: **crystalline lens (eye)**; system: visual/ophthalmic. Secondary: erythrocyte membrane (hematologic, serologic only).
- **Tissue / cell level.** Lens epithelial cells and lens fiber cells (epithelial-derived); erythrocytes. UBERON:0000965 (lens); CL:0011004 (lens fiber cell); CL:0000232 (erythrocyte).

- **Subcellular level. Golgi apparatus** (site of glycosyltransferase activity; GO:0000139 Golgi membrane); cell-surface/membrane glycoproteins and glycolipids.
- **Localization / laterality. Bilateral** congenital cataract.

8. Temporal Development

- **Onset.** Cataract is **congenital** (present at birth). The adult i phenotype reflects failure to switch from fetal i to adult I glycans, normally completed within ~18 months of life.
- **Progression.** Lens opacity itself is structurally static, but **visual deprivation is progressive** without early intervention (amblyopia). The blood-group trait is **stable and lifelong**.
- **Critical period.** The **first months of life** are the intervention window — early cataract extraction is essential to prevent irreversible amblyopia.

9. Inheritance and Population

- **Inheritance.** Autosomal recessive (biallelic *GCNT2* LOF).
- **Penetrance / expressivity.** Cataract penetrance is high for shared-exon/whole-gene genotypes; **expressivity is variable** (cataract morphology varies; the i–cataract coupling depends on isoform, Finding 2).
- **Consanguinity / founder effects.** Strong role for **consanguinity**; described in Pakistani, Arab, and Japanese pedigrees. Population-specific allelic effects underlie the East-Asian-vs-White difference in i–cataract coupling (Finding 3).
- **Epidemiology.** Congenital cataract overall affects **3–6 per 10,000 live births** ([PMID: 27609212](#)); *GCNT2*-related cases are a small, rare subset. Precise CTRCT13 prevalence is not established.
- **Sex ratio.** No sex bias expected (autosomal).
- **Carrier frequency.** Not precisely established; higher effective carrier burden in consanguineous communities.

10. Diagnostics

- **Clinical.** Slit-lamp ophthalmologic exam identifies bilateral congenital cataract. **Blood-group serology** detects the adult i phenotype (anti-I reactivity pattern).
- **Genetic testing (recommended approach).**
- **Single-gene / panel:** *GCNT2* sequencing; congenital-cataract gene panels (e.g., 51-gene pediatric cataract panels; [PMID: 28839118](#)).

- **WES/WGS:** effective, especially with **CNV/deletion-aware analysis** — several causal alleles are large deletions ([PMID: 27936067](#), [PMID: 21761136](#)).
- **CMA / MLPA / deletion assays:** important because **standard PCR/exon sequencing can fail** over deleted regions; failed amplification is itself a clue.
- **Differential diagnosis.** Other genetic congenital cataracts (crystallins *CRYAA/CRYAB/CRYGC/CRYGD*, connexins *GJA3/GJA8*, *MIP/AQP0*, *HSF4*, *BFSP1/2*); syndromic causes (e.g., *TFAP2A/BOFS* if deletion extends toward *TFAP2A*); metabolic cataract (galactosemia). The **adult i blood group** is a distinctive clue pointing to *GCNT2*.

11. Outcome / Prognosis

- **Survival / life expectancy. Normal** — CTRCT13 is not life-limiting. The adult i phenotype is benign (matters chiefly for transfusion compatibility).
- **Morbidity / function.** Driven by cataract-related visual outcomes. With early surgery and rehabilitation, functional vision is generally achievable; without it, **deprivation amblyopia** and blindness result.
- **Post-surgical complications (general pediatric cataract literature).** Posterior capsule opacification, secondary glaucoma, and post-operative high myopia are recognized long-term risks after congenital cataract surgery ([PMID: 41276154](#); [PMID: 41419074](#); [PMID: 42288323](#)) — these inform post-operative surveillance rather than the primary disease.
- **Prognostic factors.** Timing of surgery, laterality, associated ocular anomalies (e.g., microcornea), and amblyopia management.

12. Treatment

- **Definitive treatment. Surgical cataract extraction** (lensectomy) with **intraocular lens implantation or aphakic correction**, followed by **amblyopia therapy** and refractive rehabilitation. NCIT: cataract surgery / lens extraction; intraocular lens implantation.
- **Pharmacotherapy / advanced therapeutics. None disease-specific.** There is no gene, cell, RNA, or small-molecule therapy correcting the *GCNT2* glycosylation defect. Management is symptomatic/surgical.
- **Supportive.** Low-vision aids, optical correction, and long-term ophthalmologic follow-up for post-operative complications (PCO, glaucoma, myopic shift).
- **Personalized medicine.** Genotype (isoform affected) informs counseling regarding the blood-group trait and recurrence risk, but does not currently alter cataract management.

13. Prevention

- **Primary prevention.** Not possible for the monogenic disease; **genetic counseling** for at-risk consanguineous families is the main tool.
- **Secondary prevention. Early detection** — newborn red-reflex screening enables timely surgery within the critical amblyogenic window.
- **Reproductive options.** Carrier testing, **prenatal diagnosis**, and **preimplantation genetic testing** are available for families with a known *GCNT2* genotype.
- **Counseling.** Autosomal-recessive recurrence risk (25% for carrier × carrier couples); emphasize benign nature of the blood-group trait but the importance of early cataract intervention.

14. Other Species / Natural Disease

- **Taxonomy / orthologs.** *GCNT2* orthologs exist across mammals (human *GCNT2*, NCBI Gene 2651; mouse *Gcnt2*). The I-branching enzyme and I/i antigen system are evolutionarily conserved.
- **Natural disease in other species.** No well-characterized naturally occurring *GCNT2*-cataract analog was identified in the reviewed literature (OMIA/veterinary relevance not established here).
- **Comparative biology.** The biochemistry (polyLacNAc branching) is conserved; the enzyme's role has been studied in the context of cancer glycobiology across systems.

15. Model Organisms

- **Available systems.** No dedicated CTRCT13 disease model was identified in the reviewed literature. Relevant experimental systems include **in-vitro enzymology and glycan chemistry** (PMID: 37909475) and **cell-line studies of GCNT2/I-branching** in cancer biology (e.g., melanoma models showing *GCNT2* loss enhances growth, PMID: 30135430).
- **Recapitulation / limitations.** Existing models illuminate **enzyme function and glycan biology** but do **not** reproduce the lens-opacity phenotype; a lens-specific *Gcnt2*-null model would be needed to test the inferred cataract mechanism (Step 5b).
- **Applications.** Enzyme mechanism, substrate specificity, isoform biology, and (separately) the role of I-branching in malignant progression.

Evidence Base

PMID	Title (abbrev.)	Contribution
21761136	<i>Alu repeat-mediated GCNT2 deletion...</i>	Identifies GCNT2; 93 kb Alu-mediated deletion; SINE-rich locus (F001, F004)
15161861	<i>Nonsense mutation in GCNT2...</i>	LOD 8.75; W326X/W328X across all isoforms; shared-exon lesion → both traits (F001)
12424189	<i>Molecular genetics of the human I locus...</i>	Three-isoform architecture; defines i→I conversion reaction (F002, F005)
11739194	<i>Molecular basis of the adult i phenotype...</i>	IGnT = I gene; Gly348Glu/Arg383His abolish activity; Asian cataract association (F001, F005)
17076854	<i>Novel IGnT allele...</i>	Exon-1C variant → adult i without cataract; proves isoform dissociation (F002)
27609212	<i>Homozygous deletion of GCNT2 A/B first exons...</i>	GCNT2B = lens isoform, GCNT2C = RBC isoform; ~98 kb Alu-NHEJ deletion (F002, F004)
41813616	<i>Biallelic GCNT2 variants implicate exon 1B...</i>	Exon-1B frameshift + 75 kb CNV; exon 1B clinically relevant for cataract (F002, F004)
27936067	<i>Deletion at GCNT2 causes arCC...</i>	~190 kb indel; LOD 5.78; failed exon PCR flags structural deletion (F004)
432928	<i>Phenotype i with congenital cataract in Japanese</i>	17/18 i-individuals with cataract; strong East Asian coupling (F003)
3799539	<i>Reduced Ii-ataract association in Whites</i>	3/10, and 0/31; weak coupling in Whites (F003)
24105809	<i>β3GnT2 and GCNT2 co-regulate polyLacNAc</i>	GCNT2 = β1,6-branching enzyme; cooperates with β3GnT2 (F005)
37909475	<i>Chemo-enzymatic synthesis of I-branched polyLacNAc</i>	Confirms GCNT2 branching chemistry (mechanism support)
28839118	<i>High-throughput pediatric cataract screening</i>	Panel-based diagnostic context
30135430 / 31213534	<i>GCNT2/I-branching in cancer</i>	I-branching biology in other tissues (comparative/model context)

Evidence-source types: Human clinical/pedigree (F001–F004; PMIDs 21761136, 15161861, 27609212, 27936067, 432928, 3799539, 17076854, 41813616, 11739194); in-vitro/enzymatic and glycochemistry (F005; PMIDs 12424189, 24105809, 37909475); cell-line/comparative (30135430, 31213534).

Limitations and Knowledge Gaps

1. **Lens biophysical mechanism unproven.** The step from "unbranched polyLacNAc on lens cells" to "lens opacity" (Step 5b) is **inferred from enzyme chemistry**, not demonstrated in lens tissue. The specific lens glycoprotein target(s) are unknown.
2. **No animal model of the cataract.** Existing GCNT2 models address enzyme/cancer biology, not lens phenotype; phenotype recapitulation is untested.
3. **Epidemiology imprecise.** Exact CTRCT13 prevalence, carrier frequency, and gnomAD allele frequencies are not established; most data derive from small consanguineous pedigrees.
4. **Isoform–phenotype map still coarse.** The lens-relevant transcript is being refined (exon 1B evidence is recent, [PMID: 41813616](#)); the precise transcript requirements for cataract vs. blood group are not fully resolved.
5. **TFAP2A contiguous-deletion risk uncharacterized.** Large deletions extending toward *TFAP2A* have unclear phenotypic consequences (possible BOFS overlap), not systematically studied.
6. **Population-specific allelic architecture** underlying the East-Asian-vs-White coupling difference is hypothesized from phenotype data, not directly demonstrated by allele-frequency comparison.

Proposed Follow-up Experiments / Actions

1. **Lens-specific *Gcnt2* isoform knockout mouse** (or human lens organoid / iPSC-derived lens model) to test whether loss of the GCNT2B/shared-exon transcript is sufficient to produce cataract, and to identify the glycoprotein substrate driving opacity.
2. **Glycoproteomic profiling of lens tissue** (PRIDE/ProteomeXchange workflow) comparing GCNT2-deficient vs. control lens to map altered polyLacNAc branching on specific lens proteins.

3. **Systematic CNV survey of unsolved recessive congenital cataract** cohorts at 6p24 with deletion-aware WES/WGS + MLPA, to quantify the true contribution of structural *GCNT2* alleles.
 4. **Allele-frequency/haplotype comparison** across East Asian vs. European cohorts to formally test whether shared-exon vs. isoform-restricted alleles explain the population difference in i-cataract coupling.
 5. **Functional assays of VUS** (enzyme-activity assays for missense variants, à la Gly348Glu/Arg383His) to enable ACMG PS3-level classification.
 6. **Curate CTRCT13-specific epidemiology** from consanguineous-population registries to establish prevalence and carrier frequency.
 7. **Characterize the *GCNT2-TFAP2A* deletion breakpoint landscape** to define genotype–phenotype rules for contiguous deletions and any BOFS overlap risk.
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Report compiled from a 5-iteration autonomous investigation; 5 confirmed findings, 21 papers reviewed. All mechanistic and clinical claims are cited to primary literature (PMID). Evidence-source types are distinguished as human clinical, in-vitro/enzymatic, or comparative/model as noted.
